

Replacing Imported Waste Paper with an All-India Fibre Blend

Cutting Indian kraft paper mills' dependence on US/EU imported fibre for high-BF grades — matching strength, RCT, colour and smell at roughly half the cost, sourced entirely in India, with no machine change.

Prepared 2 October 2026 · Opus 4.8 research

Scope: problem, solution, economics, business model, competition, mill-trial protocol & sourcing

All prices indicative public data (India 2025–26) — confirm by quotation before acting.

Contents

1 · Executive summary	below
2 · The problem, decoded	p.2
3 · Market & economics	p.3
4 · The committed India solution	p.4
5 · Does it match US quality?	p.5
6 · Business model	p.6
7 · Who is already doing this in India	p.7
8 · Mill-trial protocol & sourcing	p.8
9 · Risks, caveats & next steps	p.9
Appendix · Glossary & sources	p.10

1 · Executive summary

Indian mills make 16–18 BF kraft easily from local recycled waste. For 22 BF and above — and for **RCT (edge-crush strength)**, not just burst — they must blend in **imported US/EU waste paper** that carries long virgin fibre India cannot produce. That import is the pain: volatile ocean freight, a ~2-month price lock-in, and ~₹25–30/kg landed cost.

The honest finding. No single cheap Indian material is half the price of US waste on a like-for-like tonne. The real ~50% saving comes from an **all-India blend** whose cost to hit the same BF + RCT is about half: a cheap domestic base + a small share of domestic long fibre (jute waste, kraft trim, bamboo) + strength chemistry that multiplies each fibre's strength — dropped in where the import went, **no machine change**.

5.76 Mt
India recovered-paper imports
Jan–Oct 2025 (+19% YoY)

~50%
Target cut in furnish cost to
reach the same grade

500+
Addressable kraft mills (50
already associated)

Open
The trading model (stocked
blend for small mills) looks
unoccupied

This dossier is a researched proposal, not a lab result. The one true proof is a 4–6 week mill trial (section 8).

2 · The problem, decoded

PROBLEM

Recycled Indian OCC is short-fibre and over-recycled, so it caps out around 18 BF. Climbing the BF ladder — and getting compression strength — needs long virgin fibre, imported because India has no fresh wood-pulp capacity (forest limits).

GRADE	HOW IT IS MADE TODAY	IMPORTED-FIBRE DEPENDENCE
16–18 BF	100% Indian OCC + starch	None
20–24 BF	Indian OCC + 30–50% imported	Rising
28–35 BF	Import-heavy blend	High
40–45 BF	Premium US DLK/DSOCC	Locked to imports

The technical crux: starch buys burst, not compression

PROPERTY	WHAT DRIVES IT	STARCH / BONDING	LONG IMPORTED FIBRE
BF (Burst Factor)	Inter-fibre bonding	High ✓	High ✓
RCT (Ring Crush)	Fibre network / stiffness	Low ✗	High ✓

The import buys two things at once — long fibre for RCT *and* bonding surface for BF. Starch replaces only the second. Any real substitute must rebuild the **compression-bearing fibre network**, which is the hard, valuable part.

The four pains: volatile ocean freight (Red Sea) · ~2-month shipment lead with price lock-in · US\$150–250/t fibre · ~10% furnish yield loss. The durable pain is lead-time + working capital + margin squeeze in an over-supplied market — not freight alone.

METRIC	FIGURE (2025–26)	SOURCE
India recovered-paper imports (Jan–Oct 2025)	~5.76 Mt, +19.3% YoY	GTAIC
Average import proxy price (LTM)	~US\$200/t	GTAIC
US vs Italy proxy ("barbell")	US\$216.7/t vs US\$175.2/t	GTAIC
EU OCC / US DSOCC peaks	~US\$200 / ~US\$250/t	Paswara (P&P Times)
Recent EOCC 95/5	~US\$150 CNF/t	Fastmarkets
Landed imported fibre (per voice-note & market)	~₹25–30/kg	Market / brief
Domestic kraft market	Over-capacity; 5-day/month mill shutdowns	P&P Times

Imported fibre lands ~₹25–30/kg delivered, matching the field brief. Freight is **volatile, not permanently high** — so the pitch should anchor on lead-time, working capital and ₹/tonne, not on "freight always rises."

4 · The committed India solution

SOLUTION

An all-India long-fibre blend, dropped in exactly where US waste went — no machine change at the mill.



Where the 50% actually comes from

Not one fibre cheaper than US waste (virgin bamboo pulp ≈ ₹58/kg is dearer). The saving is on the **blended cost of hitting the grade**: replace expensive imported waste with a cheap base + a small long-fibre share + chemistry that lets less long fibre do the work.

LAYER	INDIA SOURCE	ROLE	INDICATIVE COST
Base 60–70%	Domestic OCC + bagasse/wheat straw	Bulk furnish	₹10–15/kg
Long fibre 15–25%	Jute waste / gunny · domestic kraft trim · bamboo (premium)	Delivers RCT	₹8–22/kg
Multiplier	GPAM + PVAm + starch + enzyme	More strength per fibre	+₹1–2.5/kg paper
= Blend	—	Same BF/RCT, India-only	₹12–18/kg (~40–50% less)

Why it genuinely works, not just a cost trick

- **Jute / recovered gunny** — long, strong bast fibre; documented as able to replace long-fibre bamboo and imported softwood; can repulp in the mill's existing hydrapulper (drop-in, no new plant — validate smell/cleaning).
- **Bamboo** — 2–3 mm fibre ≈ softwood → full RCT; and the **legal unlock**: since the 2017 Forest Act amendment, bamboo on non-forest land is not a "tree" (no permit, freely traded, government-promoted). Buy market pulp or toll-pulp; don't build a mill.
- **Domestic kraft trim / DLK** — imported-grade long fibre, generated in India, no sea freight.
- **Chemistry + enzyme** — "raw materials you add" at the existing wet-end/refiner.

5 · Does it match US quality?

Researched confidence per attribute. The only true proof is the trial in section 8.

US MATERIAL GIVES	CAN INDIA BLEND MATCH?	CONFIDENCE	WHAT TO WATCH
BF (burst)	Yes	High	Starch + GPAM handle it
RCT (strength)	Yes, to a point	Med-High	35–45 BF needs more bamboo → cost gap narrows
Colour	Yes	High	All browns; minor dye tuning
Smell	Mostly — must verify	Medium	#1 trial checkpoint (jute/gunny odour)
Runs on same machine	Yes, with enzyme	Med-High	Confirm drainage & speed hold

Honest: strength and colour are the safe parts; smell and drainage are the real watch-items — which is exactly why they are explicit pass/fail lines in the trial protocol. No blind guarantee; expect 1–2 dose iterations.

Value proposition: Cut imported-fibre dependence 30–60% at a net cost saving — same paper, no machine change, stocked locally and priced daily, zero 2-month shipment lock-in.

REVENUE STREAMS

- Gain-share on the strength kit (₹/tonne paper)
- Trading margin on domestic long fibre
- Recurring re-supply + recipe service

PRICING — GAIN-SHARE

- Split the saving: mill keeps ~half, you keep ~half
- Priced per tonne improved, not per kg additive
- Daily index pricing — no booking risk

CUSTOMER SEGMENTS

- Beachhead: 50 associated mills
- Core: small/mid kraft mills — Gujarat (Morbi/Vapi/Ahmedabad)
- Scale: 500+ pan-India

KEY PARTNERS

- Chemical makers (Solenis/BASF, Kemira, Ecolab, Buckman)
- Enzyme (Advanced Enzymes, Novonesis)
- Box plants & jute mills (fibre)

Unit economics — illustrative, per tonne of 28 BF paper (imports 40% → 20%)

LINE	VALUE
Raw-material saving (imported → domestic swap)	+₹1.60/kg paper
Added kit cost (chemistry + enzyme)	−₹1.00/kg paper
Gross benefit pool to split	₹0.60/kg paper
Your captured share (illustrative ₹0.5–1.0/kg)	~₹0.25–0.5 Cr/month per 500 t/day mill

GO-TO-MARKET

PHASE 0 · 0–2 MO	PHASE 1 · 2–6 MO	PHASE 2 · 6–18 MO	PHASE 3 · 18 MO+
One mill trial → prove ₹/t saved & % imports cut	Lock formulation, file patent, Gujarat warehouse, first 5–10 mills	50 associated mills + Gujarat belt; recipe library by grade	Pan-India 500+; add MFC premium tier as price falls

MOAT

Patented recipe + per-BF dosing method · local stock (daily price, no lag) · first-mover channel into under-served small mills · gain-share pricing aligns incentives.

Reality check: figures are illustrative and need live-mill validation. Pitch "cut imports 30–60% and de-risk cash flow," never "eliminate imports at half price."

7 · Who is already doing this in India

COMPETITION

Idea active, model open. Domestic-fibre substitution is already pursued — but as **integrated mills** making their own paper, or **chemical majors** selling to large mills. Nobody appears to sell a stocked, drop-in domestic long-fibre + chemistry blend to small kraft mills on gain-share.

PLAYER	FEEDSTOCK	MODEL	OVERLAP	THREAT
Yash Pakka (Ayodhya)	Bagasse (~75k t/yr)	Integrated; CHUK tableware, exports 43 countries	Agro, food/compostable focus	Med
Ruchira Papers (HP)	Wheat straw, bagasse, rice husk	Integrated agro packaging grades	Sells finished paper	Med
KR Papers (Gujarat)	Bagasse + agro	Integrated kraft/packaging, 30+ yr	Agro kraft, your region	Med
NatureWrks (IIT-M)	Ag waste	Start-up, own products	Different class	Low
Amazon × IIT-Roorkee	Farm waste	R&D, captive (~2027)	Not merchant fibre	Low
Bamboo missions / DPRs	Bamboo	Proposed 20–40k MT plants	Supply source & future rival	Med if built
Solenis/Kemira/Ecolab/Buckman	Chemistry	Dry-strength to large mills	No fibre bundle / small-mill model	Low

WHAT'S TAKEN

- Integrated agro/bamboo mills
- Chemical majors → large mills
- Government/research pushing substitution

WHAT'S OPEN (YOUR SLOT)

- A trader, not a mill
- Drop-in blend sold TO small mills
- Warehouse-stocked, daily-priced
- Gain-share, no machine change

Caveat: absence of public evidence ≠ nobody private. Scout 5–10 mills + suppliers before scaling.

8 · Mill-trial protocol & sourcing

PROOF

Prove it on one mill in ~4–6 weeks: run CONTROL (current imported recipe) vs 3 India blends; measure the same properties; the cheapest blend that passes all is the product.

RECIPES (% OF OVEN-DRY FURNISH)

COMPONENT	T1 DROP-IN	T2 +BAMBOO
Domestic OCC	70	60
Imported fibre	0	0
Bagasse	–	10
Jute waste / gunny	15	10
Domestic kraft trim	15	10
Bamboo pulp	–	10
Starch / GPAM / resin	1.0 / 0.2 / 0.3%	1.0 / 0.2 / 0.3%
Enzyme (refiner)	0.3 kg/t	0.3 kg/t

TESTS & PASS TARGETS

PROPERTY	METHOD	PASS VS CONTROL
BF	IS 1060 / ISO 2758 / T403	Within ±1 BF point
RCT	ISO 12192 / TAPPI T822	≥ 95% of control
Cobb	ISO 535	Within ±10%
Colour	ΔE / light-box	Visually matched
Smell	5-person blind panel	Neutral / no off-odour
Runnability	Drainage, speed, breaks	Speed ≥ 95%, no new breaks

SOURCING (INDICATIVE 2025 — CONFIRM BY QUOTE)

MATERIAL	WHERE / WHO	PRICE
Domestic OCC	Local traders; Gujarat belt	₹14–23/kg
Jute waste / gunny (not raw jute)	WB jute mills; scrap dealers	gunny scrap ~₹15–17/pc
Domestic kraft trim / DLK	Box plants / converters	₹18–24/kg
Bagasse	Sugar mills (MH/UP/GJ)	₹1.4–2.2/kg
Bamboo unbleached pulp	Buy market / toll-pulp	₹58–62/kg
Starch	Gujarat Ambuja, Sukhjit, Roquette	₹45–70/kg
GPAM / resin	Anmol Polymers, UP Alums, majors	quote
Enzyme	Advanced Enzyme Technologies, Novonesis	quote

Risks & honest caveats

- **No blind guarantee** — results depend on the mill's furnish, water and machine; expect 1–2 dose iterations.
- **Highest grades (35–45 BF)** lean more on bamboo → cost gap narrows; prove the 18–28 BF sweet spot first.
- **Smell & drainage** are the real watch-items with jute/gunny & agro.
- **Chemistry is partly commoditised** — the moat is IP + local stock + service + the small-mill channel, not the molecules.
- **Credible incumbents** (Yash Pakka, Ruchira, KR, bamboo plants) could move down-market; speed + patent matter.
- **All ₹ figures indicative** — validate by quotation and on a live mill before committing capital.

Immediate next steps

- Pick one associated mill; capture its current BF, RCT, furnish %, imported-fibre cost, starch dose.
- Get trial samples: chemistry/enzyme kits + a few hundred kg of each Indian fibre.
- Run the 4–6 week trial (section 8); measure ₹/tonne saved & % imports displaced.
- If net-positive: lock the formulation, file IP, set up the Gujarat warehouse, start gain-share sales.
- In parallel, scout 5–10 mills + suppliers to confirm the model is still open.

GLOSSARY

BF Burst Factor (bonding-driven) · **RCT** Ring Crush Test, edge compression / stacking strength (fibre-network-driven) · **CMT** Concora fluting crush · **OCC** Old Corrugated Containers · **DLK/DSOCC** premium long-fibre imported grades · **GPAM** glyoxalated polyacrylamide · **PVAm** polyvinylamine · **MFC/NFC** micro/nano-fibrillated cellulose · **Hornification** irreversible fibre bonding loss on recycling · **Yield loss** input vs output gap (~8–12%).

SOURCES

GTAIC India waste-paper market 2025–26 — gtaic.ai · Pulp & Paper Times (excess capacity; Paswara OCC prices) — thepulpandpapertimes.com · Fastmarkets recovered paper — fastmarkets.com · Xeneta Red Sea / freight outlook — xeneta.com · BioResources nanocellulose/CPAM on recycled board — bioresources.cnr.ncsu.edu · Ecolab/Nalco dry-strength (GPAM×starch ring-crush) · Solenis dry-strength case history — solenis.com · TAPPI enzyme refiner mill trials — tappi.org · NITI Aayog bamboo for pulp — niti.gov.in · BioResources ICAR-CIRCOT MFC cotton linters · PIB 2017 Forest Act amendment (bamboo not a "tree") — pib.gov.in · IPPTA economics of bamboo & "Can we use Jute" — ippta.co · IndexBox bamboo pulp price India · ISO 12192 / TAPPI T822 (RCT) — zwickroell.com · Advanced Enzyme Technologies — advancedenzymes.com · paperindex.com (chemical & waste-paper directories) · krpapers.com · aboutamazon.com (Amazon × IIT-Roorkee) · IIT-Madras NatureWrks release · IPMA — ipmaindia.org. Full clickable URLs in research/kraft-paper-import-substitution.md (Appendix B).

End of dossier · Prepared 2 Oct 2026 · All figures indicative public data for planning, not firm quotes.